Chapter 4

Multivariate Random Variables

In this chapter, we will review some topics related to random vectors, which will be of use in the following chapters.

4.1 Random vectors

A **random vector** is a vector of random variables.[1] Consider the random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_m \end{pmatrix}, \quad \boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}. \tag{4.1}$$

The **expected value** of $\boldsymbol{X}$ is

$$\mathbb{E}\,\boldsymbol{X} = \begin{pmatrix} \mathbb{E}\,X_1 \\ \vdots \\ \mathbb{E}\,X_m \end{pmatrix}. \tag{4.2}$$

The **correlation matrix** of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the $m \times n$ matrix $\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]$, whose i, jth element is $\mathbb{E}[X_i Y_j]$. The **cross-covariance matrix** $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})$ of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the matrix $\mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T]$, whose i, jth element is $\mathrm{Cov}(X_i, Y_j)$. The covariance of a vector $\boldsymbol{X}$ is $\mathrm{Cov}(\boldsymbol{X}) = \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X})$. The **conditional expectation** $\mathbb{E}[\boldsymbol{X}|\boldsymbol{Y}]$ of $\boldsymbol{X}$ given $\boldsymbol{Y}$ is a vector whose ith element is $\mathbb{E}[X_i|\boldsymbol{Y}]$.

If the elements of $\boldsymbol{X}$ are uncorrelated, then $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$ and the covariance matrix becomes diagonal. If, in addition, $\mathrm{Cov}(X_i, X_i) = \mathrm{Var}(X_i) = \sigma^2$, i.e., all elements of $\boldsymbol{X}$ have the same variance σ^2, then $\mathrm{Cov}(\boldsymbol{X}) = \sigma^2 I$.

Example 4.1 (Multinomial Distribution)**.** Consider rolling a fair six-sided die n times, with each roll being independent. Let

$$\boldsymbol{X} = (X_1, X_2, X_3, X_4, X_5, X_6)^T$$

be a random vector where X_i represents the number of times face i appears. The vector $\boldsymbol{X}$ follows a multinomial distribution with parameters n and $\boldsymbol{p} = (1/6, 1/6, 1/6, 1/6, 1/6, 1/6)^T$, where p_i is the probability of face i appearing on a single roll.

Since the faces must sum to n rolls, we have the constraint $\sum_{i=1}^{6} X_i = n$, which means the components

[1] We use lowercase bold letters to denote deterministic vectors, uppercase bold letters to denote random vectors, and uppercase sans serif letters, such as A, to denote matrices.

cannot be independent. For a fair die:

$$\mathbb{E}[X_i] = \frac{n}{6} \quad \text{for all } i, \tag{4.3}$$

$$\mathrm{Var}(X_i) = n \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{5n}{36}, \tag{4.4}$$

$$\mathrm{Cov}(X_i, X_j) = -n \cdot \frac{1}{6} \cdot \frac{1}{6} = -\frac{n}{36} \quad \text{for } i \neq j. \tag{4.5}$$

Hence, we have

$$\mathbb{E}[\boldsymbol{X}] = \frac{n}{6}\begin{pmatrix}1\\1\\1\\1\\1\\1\end{pmatrix}, \qquad \mathrm{Cov}(\boldsymbol{X}) = \frac{n}{36}\begin{pmatrix} 5 & -1 & -1 & -1 & -1 & -1 \\ -1 & 5 & -1 & -1 & -1 & -1 \\ -1 & -1 & 5 & -1 & -1 & -1 \\ -1 & -1 & -1 & 5 & -1 & -1 \\ -1 & -1 & -1 & -1 & 5 & -1 \\ -1 & -1 & -1 & -1 & -1 & 5 \end{pmatrix}. \tag{4.6}$$

Note that the off-diagonal covariances are *negative*: if we observe more occurrences of one face, there must be fewer rolls available for the other faces. This is a natural example of negative correlation arising from a constraint. △

4.1.1 Properties of expectation and covariance

For deterministic matrices A, B, deterministic vectors $\boldsymbol{a}, \boldsymbol{b}$, and random vectors $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{W}, \boldsymbol{Z}$, we have [1]

1. $\mathbb{E}[\mathsf{A}\boldsymbol{X} + \boldsymbol{a}] = \mathsf{A}\,\mathbb{E}\,\boldsymbol{X} + \boldsymbol{a}$
2. $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) = \mathbb{E}[\boldsymbol{X}(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T] = \mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})\boldsymbol{Y}^T] = \mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T] - \mathbb{E}\,\boldsymbol{X}\,\mathbb{E}\,\boldsymbol{Y}^T$
3. $\mathbb{E}[(\mathsf{A}\boldsymbol{X})(\mathsf{B}\boldsymbol{Y})^T] = \mathsf{A}\,\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]\mathsf{B}^T$
4. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}, \mathsf{B}\boldsymbol{Y} + \boldsymbol{b}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})\mathsf{B}^T$
5. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T$
6. $\mathrm{Cov}(\boldsymbol{W} + \boldsymbol{X}, \boldsymbol{Y} + \boldsymbol{Z}) = \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Z}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Z})$

Example 4.2. For a random vector $\boldsymbol{X}$ and constants $a, \boldsymbol{b}$, from property 5, we have $\mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}) = a^2\,\mathrm{Cov}(\boldsymbol{X})$. We also prove this using the other properties. The relevant properties are given in each step.

$$\begin{aligned} \mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}) &= \mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}, a\boldsymbol{X} + \boldsymbol{b}) && (4.7)\\ &\overset{6}{=} \mathrm{Cov}(a\boldsymbol{X}, a\boldsymbol{X}) + \mathrm{Cov}(a\boldsymbol{X}, \boldsymbol{b}) + \mathrm{Cov}(\boldsymbol{b}, a\boldsymbol{X}) + \mathrm{Cov}(\boldsymbol{b}, \boldsymbol{b}) && (4.8)\\ &\overset{2}{=} a^2\,\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X}) + 0 + 0 + 0 && (4.9) \end{aligned}$$

△

Example 4.3. Continuing Example 4.1, let us consider a linear transformation that counts specific types of outcomes. Define:

- Y_1 = number of prime outcomes (faces showing 2, 3, or 5)
- Y_2 = number of odd outcomes (faces showing 1, 3, or 5)

We can express $\boldsymbol{Y} = (Y_1, Y_2)^T$ as a linear transformation of $\boldsymbol{X}$:

$$\boldsymbol{Y} = \mathsf{A}\boldsymbol{X}, \quad \text{where } \mathsf{A} = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}. \tag{4.10}$$

The first row of A selects faces 2, 3, and 5 (primes), while the second row selects faces 1, 3, and 5 (odd numbers). Using property 1 and property 5, we can compute:

$$\mathbb{E}[\boldsymbol{Y}] = \mathsf{A}\,\mathbb{E}[\boldsymbol{X}] = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} n/6 \\ \vdots \\ n/6 \end{pmatrix} = \begin{pmatrix} n/2 \\ n/2 \end{pmatrix}, \tag{4.11}$$

$$\mathrm{Cov}(\boldsymbol{Y}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T. \tag{4.12}$$

Computing $\mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T$:

$$\mathrm{Cov}(\boldsymbol{Y}) = \frac{n}{36}\begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 5 & -1 & -1 & -1 & -1 & -1 \\ -1 & 5 & -1 & -1 & -1 & -1 \\ -1 & -1 & 5 & -1 & -1 & -1 \\ -1 & -1 & -1 & 5 & -1 & -1 \\ -1 & -1 & -1 & -1 & 5 & -1 \\ -1 & -1 & -1 & -1 & -1 & 5 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{pmatrix} \tag{4.13}$$

$$= \frac{n}{36}\begin{pmatrix} 12 & 6 \\ 6 & 12 \end{pmatrix} = \begin{pmatrix} n/3 & n/6 \\ n/6 & n/3 \end{pmatrix}. \tag{4.14}$$

Thus, $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2) = n/3$ and $\mathrm{Cov}(Y_1, Y_2) = n/6 > 0$. The positive covariance makes sense: outcomes 3 and 5 contribute to both counts, so observing more primes tends to mean observing more odd numbers. △

4.2 Gaussian Random Vectors (Multivariate Normal Distribution)

Recall that a random variable X is Gaussian (normal) with mean μ and variance $\sigma^2 > 0$ if the pdf of X is given by

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{4.15}$$

Definition 4.4. A collection of random variables is **jointly Gaussian** if any linear combination of these variables is Gaussian. A **Gaussian random vector**, also known as a *multivariate normal vector*, is a vector whose elements are jointly Gaussian. A collection of random vectors is jointly Gaussian if the vector obtained by concatenating them is jointly Gaussian.

Example 4.5. If $\begin{pmatrix} X \\ Y \end{pmatrix}$ is a Gaussian vector, then $Z = 2X + 3Y$ is Gaussian. Furthermore,

$$\mathbb{E}[Z] = 2\,\mathbb{E}[X] + 3\,\mathbb{E}[Y], \tag{4.16}$$

$$\mathrm{Var}(Z) = \mathrm{Cov}(2X+3Y, 2X+3Y) = 4\,\mathrm{Cov}(X,X) + 12\,\mathrm{Cov}(X,Y) + 9\,\mathrm{Cov}(Y,Y) \tag{4.17}$$

$$= 4\,\mathrm{Var}(X) + 12\,\mathrm{Cov}(X,Y) + 9\,\mathrm{Var}(Y), \tag{4.18}$$

which completely characterizes the distribution of Z as $Z \sim \mathcal{N}(\mathbb{E}[Z], \mathrm{Var}(Z))$. △

For a Gaussian random vector $\boldsymbol{X}$ of dimension d, with mean $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}$ and covariance matrix $\mathsf{K} = \mathrm{Cov}(\boldsymbol{X}) = \mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^T]$, we have

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{d/2}|\mathsf{K}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\mathsf{K}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right), \tag{4.19}$$

provided that the covariance matrix is invertible.

The elements of $\boldsymbol{X}$ are **independent** if and only if the covariance matrix is diagonal.

4.2.1 Mahalanobis Distance

The **Mahalanobis distance** is a distance metric that measures the distance between a point and a distribution. For a distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix K, the Mahalanobis distance between a point $\boldsymbol{x}$ and the distribution is defined as:

$$\Delta(\boldsymbol{x}, \boldsymbol{\mu}) = \sqrt{(\boldsymbol{x} - \boldsymbol{\mu})^T \mathsf{K}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})} \tag{4.20}$$

With this definition, the Gaussian pdf in (4.19) can be rewritten as

$$\log p_{\boldsymbol{X}}(\boldsymbol{x}) \doteq -\frac{1}{2}\Delta^2(\boldsymbol{x}, \boldsymbol{\mu}), \tag{4.21}$$

where $\doteq$ indicates equality up to an additive constant that does not depend on $\boldsymbol{x}$.

This distance measure differs from the Euclidean distance by accounting for the correlation structure between variables through the covariance matrix K. It is invariant under full-rank linear transformations of the data.

The Mahalanobis distance provides geometric intuition for multivariate Gaussian distributions. Points with the same probability density lie on ellipsoids of constant Mahalanobis distance from the mean. The principal axes of these ellipsoids align with the eigenvectors of the covariance matrix, and their lengths are proportional to the square roots of the eigenvalues.

Example 4.6. Recall that the **correlation coefficient** ρ between two random variables X_1 and X_2 is defined as

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}} \tag{4.22}$$

The covariance matrix of a bivariate Gaussian random vector $\boldsymbol{X} = (X_1, X_2)^T$ can be expressed in terms of the variances $\sigma_1^2 = \mathrm{Var}(X_1)$, $\sigma_2^2 = \mathrm{Var}(X_2)$, and the correlation coefficient ρ as

$$\mathsf{K} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

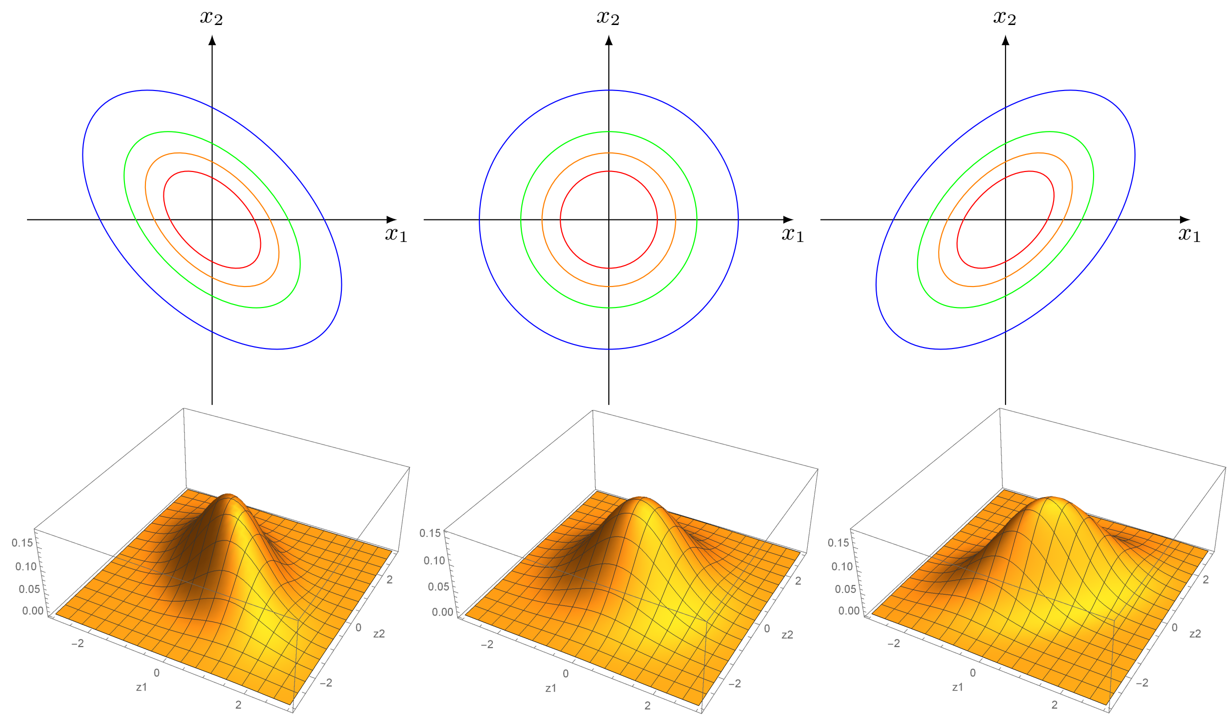
Consider three cases with $\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and variances $\sigma_1^2 = \sigma_2^2 = 1$, but different correlation coefficients:

- Case 1: $\rho = -0.5$, giving $\mathsf{K}_1 = \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix}$
- Case 2: $\rho = 0$, giving $\mathsf{K}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- Case 3: $\rho = 0.5$, giving $\mathsf{K}_3 = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

Plots for these distributions are given in Figure 4.1 illustrate how the correlation affects the shape of the distribution for a bivariate Gaussian, when the variances are the same:

- Negative correlation ($\rho < 0$): The ellipse is oriented along the line $y = -x$, indicating that as one variable increases, the other tends to decrease.
- Zero correlation ($\rho = 0$): The ellipse becomes a circle, indicating that the variables are independent.
- Positive correlation ($\rho > 0$): The ellipse is oriented along the line $y = x$, indicating that both variables tend to increase or decrease together.

$\triangle$



(a) $\rho = -0.5$ (b) $\rho = 0$ (c) $\rho = 0.5$

Figure 4.1: Contours of equal probability density and 3D pdfs for bivariate Gaussian distributions with the same variance but different correlation values. When $\rho < 0$, the ellipse is oriented along $y = -x$; when $\rho = 0$, we get a circle; when $\rho > 0$, the ellipse is oriented along $y = x$.

If $\mathsf{K} = \sigma^2 \mathsf{I}$ (i.e., the variables are uncorrelated with equal variances), the Mahalanobis distance is proportional to the Euclidean distance. The Mahalanobis distance can be extended to measure the **distance between two normal distributions** $\mathcal{N}(\boldsymbol{\mu}_1, \mathsf{K})$ and $\mathcal{N}(\boldsymbol{\mu}_2, \mathsf{K})$ with the same covariance:

$$\Delta(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2) = \sqrt{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathsf{K}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)} \tag{4.23}$$

These properties make the Mahalanobis distance particularly useful in multivariate analysis, pattern recognition, and classification problems where the correlation structure of the data is important.

4.2.2 Conditional distributions of a Gaussian vector

Suppose $\boldsymbol{X}$ is a d-dimensional Gaussian random vector with mean $\boldsymbol{\mu}$ and covariance matrix K, i.e., $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathsf{K})$. Partition $\boldsymbol{X}$ as

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_a \\ \boldsymbol{X}_b \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \quad \mathsf{K} = \begin{pmatrix} \mathsf{K}_{aa} & \mathsf{K}_{ab} \\ \mathsf{K}_{ba} & \mathsf{K}_{bb} \end{pmatrix},$$

where $\boldsymbol{X}_a$ is k-dimensional and $\boldsymbol{X}_b$ is $(d-k)$-dimensional.

The conditional distribution of $\boldsymbol{X}_a$ given $\boldsymbol{X}_b = \boldsymbol{x}_b$ is also Gaussian:

$$\boldsymbol{X}_a \mid \boldsymbol{X}_b = \boldsymbol{x}_b \sim \mathcal{N}(\boldsymbol{\mu}_{a|b}, \mathsf{K}_{a|b}),$$

where

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \mathsf{K}_{ab} \mathsf{K}_{bb}^{-1} (\boldsymbol{x}_b - \boldsymbol{\mu}_b), \tag{4.24}$$

$$\mathsf{K}_{a|b} = \mathsf{K}_{aa} - \mathsf{K}_{ab} \mathsf{K}_{bb}^{-1} \mathsf{K}_{ba}. \tag{4.25}$$

This result is fundamental in regression, inference, and Gaussian process modeling, as it describes how observing some components of a Gaussian vector affects the distribution of the remaining components.

Example 4.7. Let $\boldsymbol{X} = (X_1, X_2)^T \sim \mathcal{N}(\boldsymbol{\mu}, \mathsf{K})$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \mathsf{K} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

Then the conditional distribution of X_1 given $X_2 = x_2$ is

$$X_1 \mid X_2 = x_2 \sim \mathcal{N}\left(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2),\ (1-\rho^2)\sigma_1^2\right).$$

We thus have $\mathbb{E}[X_1|X_2 = x_2] = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$ and $\mathrm{Var}(X_1|X_2 = x_2) = (1-\rho^2)\sigma_1^2$. △

Example 4.8. Let us standardize $\boldsymbol{X}$ of the previous example. The standardized variables are given by

$$Z_1 = \frac{X_1 - \mu_1}{\sigma_1}, \quad Z_2 = \frac{X_2 - \mu_2}{\sigma_2}.$$

It follows that (Z_1, Z_2) are jointly Gaussian with

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right).$$

The conditional distribution of Z_1 given $Z_2 = z_2$ is

$$Z_1 \mid Z_2 = z_2 \sim \mathcal{N}(\rho z_2, 1-\rho^2).$$

We thus have $\mathbb{E}[Z_1|Z_2 = z_2] = \rho z_2$ and $\mathrm{Var}(Z_1|Z_2 = z_2) = 1-\rho^2$. This gives a clear interpretation of the correlation coefficient ρ: On average, for each unit increase in Z_2, Z_1 increases by ρ units. The variance of Z_1 given Z_2 is reduced by a factor of $1-\rho^2$, reflecting the information gained about Z_1 from knowing Z_2. △

Exercise 4.9. We said above that for every unit of increase in Z_2, Z_1 increases by ρ units on average. Does that mean that for every unit of increase in Z_1, Z_2 increases by $1/\rho$ units on average? Explain. △

4.2.3 Maximum likelihood estimation

Consider a d-dimensional random vector $\boldsymbol{X} = (X_1, \dots, X_d)$ with distribution $\mathcal{N}(\boldsymbol{\theta}^*, \mathsf{K}^*)$ given in (4.19), where $\boldsymbol{\theta}^*, \mathsf{K}^*$ are unknown. Suppose we are interested in the relationship between X_d and $X_1, \dots, X_{d-1}$. For example, for $\boldsymbol{X}^T = (X_1, X_2, X_3)$, X_1 and X_2 could indicate the heights of the parents and X_3 could be the height of the child. We may, for example, be interested in finding $\mathbb{E}[X_d | X_1, \dots, X_{d-1}]$, thus estimating X_d based on $X_1, \dots, X_{d-1}$. If we find the distribution, in other words, $\boldsymbol{\theta}^*, \mathsf{K}^*$, we can do so. Furthermore, the matrix K^* can indicate which dimensions are more strongly correlated.

Consider a set of n iid samples $\mathcal{D} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_n\}$, where each $\boldsymbol{x}_i$ is a sample of $\boldsymbol{X}$. We denote the elements of $\boldsymbol{x}_i$ as $\boldsymbol{x}_i = (x_{i1}, \dots, x_{id})$.

To estimate $\boldsymbol{\theta}^*$ and K^*, we write

$$\ell(\boldsymbol{\theta}, \mathsf{K}) = \ln p(\mathcal{D}; \boldsymbol{\theta}, \mathsf{K}) = \sum_{i=1}^{n} \ln p(\boldsymbol{x}_i; \boldsymbol{\theta}, \mathsf{K}) \tag{4.26}$$

$$\doteq \frac{n}{2} \ln |\mathsf{K}^{-1}| - \frac{1}{2} \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1} (\boldsymbol{x}_i - \boldsymbol{\theta}), \tag{4.27}$$

where we have used the fact that $|\mathsf{K}^{-1}| = \frac{1}{|\mathsf{K}|}$.

As seen in the appendix (last chapter), for a symmetric matrix A, we have $\frac{d}{d\boldsymbol{v}}(\boldsymbol{y}^T \mathsf{A} \boldsymbol{y}) = 2\boldsymbol{y}^T \mathsf{A} \frac{d\boldsymbol{y}}{d\boldsymbol{v}}$. Hence,

$$\frac{\partial \ell}{\partial \boldsymbol{\theta}} = -\frac{1}{2} \sum_{i=1}^{n} 2(\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1} (-\mathsf{I}) = \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1}. \tag{4.28}$$

Setting this equal to zero yields

$$\hat{\boldsymbol{\theta}}_{ML} = \bar{\boldsymbol{x}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i. \tag{4.29}$$

Exercise 4.10. Using the facts

$$\frac{\partial}{\partial \mathsf{A}} \boldsymbol{x}^T \mathsf{A} \boldsymbol{x} = \boldsymbol{x}\boldsymbol{x}^T, \quad \frac{\partial}{\partial \mathsf{A}} \ln |\mathsf{A}| = \mathsf{A}^{-T} \tag{4.30}$$

prove that

$$\hat{\mathsf{K}}_{ML} = \frac{1}{n} \sum_{i=1}^{n} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^T \tag{4.31}$$

△

4.2.4 Bayesian estimation

We now solve the same problem using Bayesian estimation, with the following likelihood

$$\boldsymbol{X} | \boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\Theta}, \mathsf{K}), \tag{4.32}$$

$$p(\boldsymbol{x}_1^n | \boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1} (\boldsymbol{x}_i - \boldsymbol{\theta}) \right), \tag{4.33}$$

where, for simplicity, we assume K is known and we only need to estimate $\boldsymbol{\Theta}$. As the prior, we choose

$$\boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \mathsf{S}_0) \tag{4.34}$$

$$p(\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \mathsf{S}_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) \right). \tag{4.35}$$

Hence,

$$p(\boldsymbol{\theta}|\boldsymbol{x}_1^n) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\mu}_0)^T \mathsf{S}_0^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}_0) - \frac{1}{2}\sum_{i=1}^n (\boldsymbol{x}_i-\boldsymbol{\theta})^T \mathsf{K}^{-1}(\boldsymbol{x}_i-\boldsymbol{\theta})\right). \tag{4.36}$$

The exponent in the posterior is quadratic in $\boldsymbol{\theta}$, indicating that $\boldsymbol{\Theta}$ has a Gaussian distribution. So $\boldsymbol{\Theta}|\boldsymbol{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathsf{S}_n)$, for appropriate choices of $\hat{\boldsymbol{\theta}}_n$ and S_n,

$$p(\boldsymbol{\theta}|\boldsymbol{x}_1^n) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)^T \mathsf{S}_n^{-1}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)\right). \tag{4.37}$$

To find $\hat{\boldsymbol{\theta}}_n$ and S_n, we equate (4.36) and (4.37), ignoring constant multiplicative factors, which leads to

$$(\boldsymbol{\theta}-\boldsymbol{\mu}_0)^T \mathsf{S}_0^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}_0) + \sum_{i=1}^n (\boldsymbol{\theta}-\boldsymbol{x}_i)^T \mathsf{K}^{-1}(\boldsymbol{\theta}-\boldsymbol{x}_i) \doteq (\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)^T \mathsf{S}_n^{-1}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n), \tag{4.38}$$

$$\boldsymbol{\theta}^T \mathsf{S}_0^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\boldsymbol{\theta}^T \mathsf{K}^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathsf{K}^{-1}\sum_{i=1}^n \boldsymbol{x}_i \doteq \boldsymbol{\theta}^T \mathsf{S}_n^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathsf{S}_n^{-1}\hat{\boldsymbol{\theta}}_n. \tag{4.39}$$

Here, we have used the fact that

$$(\boldsymbol{a}-\boldsymbol{b})^T \mathsf{A}(\boldsymbol{a}-\boldsymbol{b}) = \boldsymbol{a}^T\mathsf{A}\boldsymbol{a} - \boldsymbol{a}^T\mathsf{A}\boldsymbol{b} - \boldsymbol{b}^T\mathsf{A}\boldsymbol{a} + \boldsymbol{b}^T\mathsf{A}\boldsymbol{b} = \boldsymbol{a}^T\mathsf{A}\boldsymbol{a} - 2\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} + \boldsymbol{b}^T\mathsf{A}\boldsymbol{b},$$

for vectors $\boldsymbol{a}, \boldsymbol{b}$ and a symmetric matrix A. Note that $\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} = \boldsymbol{b}^T\mathsf{A}\boldsymbol{a}$, as both sides are scalars and $\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} = (\boldsymbol{a}^T\mathsf{A}\boldsymbol{b})^T = \boldsymbol{b}^T\mathsf{A}\boldsymbol{a}$.

We now collect the terms of the form $\boldsymbol{\theta}^T\mathsf{A}\boldsymbol{\theta}$,

$$\boldsymbol{\theta}^T(\mathsf{S}_0^{-1} + n\mathsf{K}^{-1})\boldsymbol{\theta} - 2\boldsymbol{\theta}^T(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + \mathsf{K}^{-1}\sum_{i=1}^n \boldsymbol{x}_i) \doteq \boldsymbol{\theta}^T \mathsf{S}_n^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathsf{S}_n^{-1}\hat{\boldsymbol{\theta}}_n, \tag{4.40}$$

leading to the following values for the parameters of the posterior distribution $\boldsymbol{\Theta}|\boldsymbol{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathsf{S}_n)$,

$$\mathsf{S}_n^{-1} = \mathsf{S}_0^{-1} + n\mathsf{K}^{-1}, \tag{4.41}$$

$$\hat{\boldsymbol{\theta}}_n = \mathsf{S}_n(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\mathsf{K}^{-1}\bar{\boldsymbol{x}}) \tag{4.42}$$

$$= (\mathsf{S}_0^{-1} + n\mathsf{K}^{-1})^{-1}(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\mathsf{K}^{-1}\bar{\boldsymbol{x}}), \tag{4.43}$$

where $\bar{\boldsymbol{x}}$ is $\sum_{i=1}^n \boldsymbol{x}_i/n$. The posterior mean, $\hat{\boldsymbol{\theta}}_n$, which we can also view as a point estimate, is the weighted average of the prior mean $\boldsymbol{\mu}_0$ and what is suggested by the data $\bar{\boldsymbol{x}}$.

Exercise 4.11. Find $\hat{\boldsymbol{\theta}}_n$ and S_n^{-1} when $\mathsf{S}_0 = s^2\mathsf{I}$ and $\mathsf{K} = \sigma^2\mathsf{I}$ and interpret the results. △